

Z-cyclic Whist Tournaments for q^2 Players: A Probabilistic Approach

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Abstract

Whether a Z-cyclic whist tournament on q^2 players exists for every prime $q \equiv 3 \pmod{4}$ is a long-standing open question; previous literature has only verified the existence of such tournaments for all $q < 5000$. This approach reframes Leonard's patterned-starter construction as a matching problem on a hypergraph. We obtain $N(q) = q^3 + q^2 + 1$ valid quadruples $[a, b; u, v]$ and show that every vertex of the hypergraph $H(q)$ has degree $O(q^2)$ while every pair of distinct vertices has codegree $O(q)$. We use the Frankl–Rödl–Pippenger theorem to prove the existence of, for large q , a collection of $\frac{q+1}{2} - o(q)$ pairwise disjoint tables $(A, B, -A, -B)$.

1 Introduction

1.1 Whist tournaments

Definition 1.1. A *whist tournament* on $4n+1$ players, written $\text{Wh}(4n+1)$, is a schedule of games, each game involving two partners playing against two other partners, such that

1. the games are arranged in $4n+1$ rounds, each consisting of n games
2. each player plays in exactly one game in all but one of the rounds
3. each player partners every other player exactly once;
4. each player opposes every other player exactly twice.

The games are represented by 4-tuples (A, B, C, D) , where partners A, C oppose partners B, D . A $\text{Wh}(4n+1)$ is said to be $\mathbb{Z}$ -cyclic if A, B, C and D are in $\mathbb{Z}_{4n+1}$ and the games of round $j+1$ can be generated by adding 1 (mod $4n+1$) to all elements of round j . We adopt the convention that the initial round, indexed with $j=0$, of a $\mathbb{Z}$ -cyclic $\text{Wh}(4n+1)$ is the one in which player 0 does not play. A $\mathbb{Z}$ -cyclic tournament is therefore determined completely by its initial round, which consists of n games.

We adopt the description of $\mathbb{Z}$ -cyclic $\text{Wh}(4n+1)$ given in [1]. The conditions for initial rounds that generate a whist tournament are phrased in terms of starters. Pairs $\{A_1, B_1\}, \dots, \{A_{2n}, B_{2n}\}$ of nonzero elements of $\mathbb{Z}_{4n+1}$ form a *starter* in $\mathbb{Z}_{4n+1}$ if

$$(\alpha) \bigcup_{i=1}^{2n} \{A_i, B_i\} = \mathbb{Z}_{4n+1} - \{0\}, \quad (\beta) \bigcup_{i=1}^{2n} \{\pm(A_i - B_i)\} = \mathbb{Z}_{4n+1} - \{0\}. \quad (1)$$

Similarly, pairs $\{A_1, B_1\}, \dots, \{A_{4n}, B_{4n}\}$ form a *2-fold starter* if

- (γ) the elements A_i, B_i are exactly the elements of $\mathbb{Z}_{4n+1} - \{0\}$, each occurring twice,
- (δ) the elements $\pm(A_i - B_i)$ are exactly the elements of $\mathbb{Z}_{4n+1} - \{0\}$, each occurring twice.

The 4-tuples (A_i, B_i, C_i, D_i) can be the initial round of $\mathbb{Z}$ -cyclic $\text{Wh}(4n+1)$ if and only if [1]

- (ϵ) the pairs $\{A_i, C_i\}, \{B_i, D_i\}$, $1 \leq i \leq n$, form a starter,
- (ζ) the pairs $\{A_i, B_i\}, \{B_i, C_i\}, \{C_i, D_i\}, \{D_i, A_i\}$, $1 \leq i \leq n$, form a 2-fold starter.

One example of a starter is the collection of sets $\{x, 49-x\}$ for $1 \leq x \leq 24$; this is a starter for $q=7$. These sets cover all of the elements of $\mathbb{Z}/7^2\mathbb{Z} - \{0\}$, and the collection of differences $\{49-2x \mid 1 \leq x \leq 24\}$ also cover the elements of $\mathbb{Z}/7^2\mathbb{Z} - \{0\}$.

One example of a 2-fold starter ($q=11$) is the collection of sets $\{g^k A_i, g^k B_i, -g^k A_i, -g^k B_i\}$ [2] where $1 \leq i \leq 6$ and k ranges from $1 \leq k \leq 5$. $g = 2^{22} \equiv 81 \pmod{11^2}$ and

$$(A_i, B_i)_{1 \leq i \leq 6} = (9, 11), (16, 17), (28, 20), (8, 92), (14, 103), (4, 83)$$

This paper considers the special case of $4n+1 = q^2$ where $q \equiv 3 \pmod{4}$ is a prime.

1.2 Coset notation

Throughout the paper, we use Leonard's construction for creating whist tournaments [3]. Every table of the initial round of $\text{Wh}(q^2)$ is of the form $(A, B, -A, -B)$, so that initializing the $(q^2 - 1)/4$ ordered pairs (A, B) constructs the whole tournament. These tournaments are Z-Cyclic patterned starters (ZCPS) in the context of [4] since all of the tables have the form $(A, B - A, -B)$. Leonard took advantage of the multiplicative symmetry of $\text{Wh}(q^2)$ to reduce the number of unique variables required to fully define a tournament. Let w be a primitive root modulo q^2 , so that w has order $q(q - 1)$ in $(\mathbb{Z}/q^2\mathbb{Z})^*$, and let

$$R = \langle w^{2q} \rangle = \{2q\text{-th powers among the units of } \mathbb{Z}/q^2\mathbb{Z}\}, \quad |R| = \frac{q-1}{2}.$$

Only $\frac{(q^2-1)/4}{|R|} = (q+1)/2$ pairs (A, B) need to be specified, as the remaining tables are formed from $(rA, rB, -rA, -rB)$ where $r \in R$. We investigate the cosets of the q -th powers. For $0 \leq j \leq q - 1$ let

$$C(j) = \{w^{qk+j} \mid 0 \leq k \leq q - 2\} \quad (2)$$

be a coset of the subgroup $C(0) = \langle w^q \rangle$ of q -th powers, with $|C(0)| = q - 1$. The q cosets $C(0), \dots, C(q - 1)$ along with the set

$$C(\infty) = \{yq \mid 1 \leq y \leq q - 1\}$$

of nonzero multiples of q partition $\mathbb{Z}/q^2\mathbb{Z} - \{0\}$ into $q + 1$ classes, each of size $q - 1$. We write

$$\mathcal{I}_q = \{0, 1, \dots, q - 1\} \cup \{\infty\}, \quad |\mathcal{I}_q| = q + 1,$$

as the set of coset indices. As per [3, 2], we record a table $(A, B, -A, -B)$ by the quadruple, or configuration,

$$[a, b; u, v] \quad \text{where} \quad A \in C(a), \quad B \in C(b), \quad A + B \in C(u), \quad A - B \in C(v), \quad (3)$$

with the indices taken in $\mathcal{I}_q$.

Definition 1.2. A *table* refers to the 4-tuple $(A, B, -A, -B)$, while *quadruple* and *configuration* and interchangeably used to refer to the 4-tuple $[a, b; u, v]$.

Leonard states three general requirements for such a tournament to exist [3]: one for tables in which neither player numbers nor differences are divisible by q , one for tables whose differences are divisible by q , and one for tables whose player numbers are divisible by q . These 3 conditions are proved to be equivalent to the existence of a collection of $(q + 1)/2$ configurations $[a, b; u, v]$ in which the indices a, b and indices u, v each perfectly cover $\mathcal{I}_q$ in Proposition 3.4. Letting ∞ be an index allows us to express Leonard's 3 requirements as a single condition, which we utilize in this paper.

1.3 The problem and the present contribution

This paper focuses on the special case when $4n + 1 = q^2$, where $q \equiv 3 \pmod{4}$ is prime. The current knowledge about this problem, reviewed in detail in Section 2, is as follows. Finizio found a Z-cyclic $\text{Wh}(7^2)$ by computer search; Leonard [3] constructed examples for $q = 7, 11, 19, 23, 31$ with the above construction and verified computationally that it succeeds for every prime $q \equiv 3 \pmod{4}$ with $q < 500$; Jones and Leonard then [2] pushed the verification to $q < 5000$. The main conjecture surrounding this problem, regarding the existence of $\text{Wh}(q^2)$, still remains open:

Conjecture 1.3 (Leonard [3]). For every prime $q \equiv 3 \pmod{4}$ with $q \geq 7$ there exists a Z-cyclic patterned-starter whist tournament on q^2 players.

Leonard reports that the number of valid tables, which represents the number of valid tournaments is 5 for $q = 11$, 183 for $q = 19$, 2,978 for $q = 23$ and 2,948,855 [3] for $q = 31$. The abundance of solutions for relatively small tournaments inspired a probabilistic approach for tackling this problem, which is the main work covered in this paper.

Section 3 demonstrates this probabilistic approach. We first show that Conjecture 1.3 is equivalent to a purely combinatorial statement involving perfect hypergraph coverings through Proposition 3.4. We then count the total number of unique configurations $[a, b; u, v]$ in Theorem 3.8 through a case-by-case analysis on the q^4 pairs (A, B) . After a degree and codegree analysis of

the hypergraph $H(q)$ (Lemma 3.11, we apply the Frankl–Rödl–Pippenger theorem to conclude that almost all of the hypergraph can be perfectly covered (Theorem 3.13). However, it does not produce a tournament as Pippenger’s theorem leaves a sublinear set of coset indices uncovered. Section 3.5 describes what needs to be done to bridge the result derived in this paper to proving Conjecture 1.3

The overall organization of the proof in Section 3 is as follows: the reformulation of the problem (Section 3.1), the exact count of valid configurations for a $\mathbb{Z}$ -cyclic $\text{Wh}(q^2)$ tournament (Section 3.2), the hypergraph and degree analysis on its vertices (Section 3.3), the application of the Frankl–Rödl–Pippenger theorem (Section 3.4), and a discussion of what remains (Section 3.5).

2 Literature Review

The relevant literature falls into three groups: the general theory of whist tournaments, the work on $\text{Wh}(q^2)$ for primes $q \equiv 3 \pmod{4}$, and the probabilistic tools for near-perfect matchings in hypergraphs that we use in Section 3.

2.1 Whist tournaments and $\mathbb{Z}$ -cyclic variants

Anderson’s article [1] gives a historical account of Whist tournaments; his articles [5, 6] give the framework for constructing Whist tournaments, including the notation for starters and 2-fold starters used in Section 1.1. For general whist tournaments without any restrictions, the existence question is resolved as $\text{Wh}(v)$ exists for every $v \equiv 0, 1, \pmod{4}$ [6]

A $\mathbb{Z}$ -cyclic $\text{Wh}(v)$ has a cyclic structure in which round $j + 1$ can be generated by adding 1 ($\text{mod } v$) to all players of round j , so that a single round determines the whole tournament. This characteristic of cyclic tournaments makes $\mathbb{Z}$ -cyclic tournaments appealing for constructions, but it also makes the question of their existence more difficult to solve as the tournaments must now be created with stronger restrictions. The values of v that possess a $\mathbb{Z}$ -cyclic $\text{Wh}(v)$ is an open question; even for a small collection of v , for example $v = q^2$ with $q \equiv 3 \pmod{4}$ prime, it is an open question [6].

Finizio [4] later introduced ZCPS-whist tournaments: $\mathbb{Z}$ -cyclic tournaments whose initial round is built on the *patterned starter*, so that every table has the form $(A, B, -A, -B)$, where $-A, -B$ are taken modulo v . This reduces the number of variables needed to specify a tournament by half, and as noted in [2], produces directed and ordered whist tournaments. Furthermore, this construction connects to triplewhist literature [7, 8, 9].

2.2 Product and extension theorems for Whist Tournaments

The reason $\text{Wh}(q^2)$ for primes $q \equiv 3 \pmod{4}$ have attracted more intention compared to other tournaments is because they are able to produce infinite families of tournaments. Finizio [10] proved that a $\mathbb{Z}$ -cyclic $\text{Wh}(q^2)$ can be extended to a $\mathbb{Z}$ -cyclic $\text{Wh}(q^2 p_1^{a_1} \cdots p_r^{a_r})$ for primes $p_i \equiv 1 \pmod{4}$. Anderson, Finizio and Odoni [11] have shown that a $\mathbb{Z}$ -cyclic $\text{Wh}(q_1 q_2^2 + 1)$ exists whenever $\mathbb{Z}$ -cyclic $\text{Wh}(q_1 + 1)$ and $\text{Wh}(q_2^2)$ exist. The effect of these product theorems stated in both [3] and [2] is that each new prime q for which a $\mathbb{Z}$ -cyclic $\text{Wh}(q^2)$ exists create infinitely many new Whist tournaments $\text{Wh}(v)$. A single value of q is equivalent to an infinite family of tournaments, which explains why computational constructions of these tournaments of have been pursued even when a general construction has not been formulated.

2.3 Whist Tournament Constructions

The first $\mathbb{Z}$ -cyclic $\text{Wh}(q^2)$ for a prime $q \equiv 3 \pmod{4}$ was found by Finizio [10] for $q = 7$, by computer search.

Leonard [3] created the framework that all subsequent work on these tournaments have used, including this paper. He reduced the problem to the selection of $(q + 1)/2$ quadruples $[a, b; u, v]$ by working with the patterned starter in $\mathbb{Z}/q^2\mathbb{Z}$ and using the multiplicative symmetry under the group of $2q$ -th powers of $(\mathbb{Z}/q^2\mathbb{Z})^*$. Leonard proved that any such selection of quadruples $[a, b; u, v]$ that satisfy the three “general requirements”, which places disjoint restrictions on player and difference indices, yields a ZCPS $\text{Wh}(q^2)$.

Leonard developed the arithmetic to rigidly and efficiently construct tournaments (collections of $(q + 1)/2$ quadruples) using his framework. He constructed tournaments for $q = 7, 11, 19, 23, 31$, and reported the number of valid $\text{Wh}(q^2)$ as 5,183,2,978, and 2,948,855 for $q = 11, 19, 23, 31$

respectively. He also confirmed the existence of at least one tournament $\text{Wh}(q^2)$ for every prime $q \equiv 3 \pmod{4}$ with $q < 500$.

Jones and Leonard [2] returned to the construction two decades later. For constructing tournaments, they replaced the exhaustive backtracking used for $q \leq 31$ due to its computational inefficiency and replaced it with a hill-climbing search of the collection of configurations $[a, b; u, v]$. Using an element of g of order $(q-1)/2$ in $(\mathbb{Z}/q^2\mathbb{Z})^*$ along with $(q+1)/2$ pairs (A_i, B_i) , the initial round is reconstructed as $[g^j A_k, g^j B_k; -g^j A_k, -g^j B_k]$. Their paper accomplishes two things. Firstly, it increases the range of verified q for which $\text{Wh}(q^2)$ exist from 500 to 5000 and secondly, provides the pairs (A, B) used to construct the tournament instead of simply verifying the existence of a valid selection of quadruples.

Every known result for $\text{Wh}(q^2)$ with $q \equiv 3 \pmod{4}$ is a computational verification for tournaments of certain sizes. None of [3] or [2] generalize to show that a tournament must exist for infinitely many q or all q , they only show that $\text{Wh}(q^2)$ exists for all $q < 5000$. However, for small values of q the solution counts are abundant with exponential-like growth [2]; this observation motivated a probabilistic approach to this problem, as the difficulty lies in the construction of a tournament rather than the existence of one.

2.4 Probabilistic tools: the Rödl nibble

The probabilistic tool we use is the “nibble” method. Rödl [12] introduced it to resolve the Erdős–Hanani conjecture on asymptotically optimal packings and coverings, by randomly selecting small batches of hyperedges so that the residual hypergraph remains approximately regular at each selection.

Frankl and Rödl [13] developed the technique for near-perfect coverings, and Pippenger and Spencer [14] obtained its current form, which is a statement about the chromatic index of nearly regular hypergraphs with small codegrees. The version of Pippenger’s theorem used in this paper for coverings is standard and is stated as Theorem 3.12 below. The conclusion of the application of this theorem is a near-perfect matching covering all but εn vertices of the hypergraph.

3 Current Progress

Throughout this section $q \equiv 3 \pmod{4}$ is a prime, w is a fixed primitive root modulo q^2 , the cosets $C(j)$ are defined as in (2), and $\mathcal{I}_q = \{0, \dots, q-1\} \cup \{\infty\}$ is the set of coset indices. We write $\log_w$ for the base w discrete logarithm in $(\mathbb{Z}/q^2\mathbb{Z})^*$, so that $\log_w(k) = \alpha \iff k \equiv w^\alpha \pmod{q^2}$; for instance if $w^{20} \equiv 7 \pmod{q^2}$ then $\log_w(7) = 20$. Only the residue of $\log_w$ modulo q will matter, since by (2)

$$x \in C(j) \iff \log_w(x) \equiv j \pmod{q} \quad (x \in (\mathbb{Z}/q^2\mathbb{Z})^*, j \in \{0, \dots, q-1\}) \quad (4)$$

Finally, let $\ell := \log_w(2) \pmod{q}$.

3.1 Problem Reformulation

We first emphasize that Conjecture 1.3 is a coverage problem. Let $R = \{w^{2qk} \mid 0 \leq k \leq \frac{q-3}{2}\}$ denote the group of $2q$ -th powers among the units of $\mathbb{Z}/q^2\mathbb{Z}$. Note that $|R| = \frac{q-1}{2}$.

Lemma 3.1. *If w is a primitive root modulo q^2 , then w is a primitive root modulo q .*

Proof. The order of w modulo q^2 is $\phi(q^2) = q(q-1)$ since w is a primitive root. Let d be the order of w modulo q , so $d \mid \phi(q)$ and thus $d \mid q-1$. Let $w^d = 1 + tq$ for some integer t . Then $w^{dq} = (1 + tq)^q \equiv 1 \pmod{q^2}$ by the binomial theorem, since every term after the first is divisible by q^2 . The order of w modulo q^2 must divide dq due to the minimality of the order, so $q^2 - q \mid dq$. Factoring out q on both sides gives $q-1 \mid d$. Combining this with $d \mid q-1$ gives $d = q-1$, making w a primitive root modulo q as desired. $\square$

Lemma 3.2. *Let $A \in \mathbb{Z}/q^2\mathbb{Z} - \{0\}$ and $A \in C(a)$ for some $a \in \mathcal{I}_q$. Then, $\{\pm rA \mid r \in R\} = C(a)$.*

Proof. Since w has order $q(q-1)$ and $\gcd(2q, q(q-1)) = q \gcd(2, q-1) = 2q$, w^{2q} has order $\frac{q(q-1)}{2q} = \frac{q-1}{2}$. Additionally, let $x = \text{ord}_{q^2}(-1) = \frac{q(q-1)}{2}$ be the order of -1 modulo q^2 . $q-1 \equiv 2 \pmod{4}$ so $2q \nmid x$ but $q \mid x$. Thus, R and $-R$ are cosets of $C(0) = \{w^k \mid 0 \leq k \leq q-2\}$ with $[C(0) : R] = 2$. Consider the case when $q \nmid A$; then, $A \in C(a) \iff A = w^{qx+a}$ for some x where $a \in \mathcal{I}_q \setminus \{\infty\}$.

$$\begin{aligned}
\{\pm rA \mid r \in R\} &= \{w^{qk}A \mid 0 \leq k \leq q-2\} \\
&= \{w^{qk+qx+a} \mid 0 \leq k \leq q-2\} \\
&= \{w^{qk+a} \mid 0 \leq k \leq q-2\} \\
&= C(a)
\end{aligned}$$

Consider the case when $q \mid A, A \in C(\infty)$; let $A = yq$ for some $1 \leq y \leq q-1$. Then,

$$\begin{aligned}
\{\pm rA \mid r \in R\} &= \{\pm yqr \mid r \in R\} \\
&= \{yqw^{qk} \mid 0 \leq k \leq q-2\} \\
&= \{yqw^k \mid 0 \leq k \leq q-2\} \\
&= \{qw^k \mid 0 \leq k \leq q-2\} \\
&= \{qk \mid 1 \leq k \leq q-1\} \\
&= C(\infty)
\end{aligned}$$

The above steps hold since w is a primitive root modulo q by Lemma 3.1 and because multiplying by y permutes the elements since $\gcd(y, q) = 1$. Therefore, $\{\pm rA \mid r \in R\} = C(a)$ where $A \in C(a)$ as desired. $\square$

Lemma 3.3.

$$\bigcup_{1 \leq i \leq \frac{q+1}{2}} C(a_i) \cup C(b_i) = \mathbb{Z}/q^2\mathbb{Z} - \{0\}$$

for coset indices $a_i, b_i \in \mathcal{I}_q$ if and only if the indices $\{a_i, b_i\}$ partition $\mathcal{I}_q$.

Proof. For $x, y \in \mathcal{I}_q$, the intersection of any 2 cosets $C(x), C(y)$ will be nonempty if and only if $x = y$. Each of the $2 \cdot \frac{q+1}{2} = q+1$ cosets has $q-1$ elements, for a total of $(q+1)(q-1) = q^2 - 1$ total elements across all cosets. If 2 cosets were identical, then the union over all cosets would not be able to contain the required $q^2 - 1$ residues modulo q^2 . Thus, the $q+1$ coset indices must be distinct; since $|\mathcal{I}_q| = q+1$, the coset indices $\{a_1, \dots, a_{\frac{q+1}{2}}, b_1, \dots, b_{\frac{q+1}{2}}\}$ are a permutation of $\mathcal{I}_q$. In this scenario, the union over all cosets would contain $q^2 - 1$ nonzero distinct elements, which must be equal to $\mathbb{Z}/q^2\mathbb{Z} - \{0\}$. Therefore, the indices $\{a_i, b_i\}$ must partition $\mathcal{I}_q$. $\square$

Proposition 3.4. *There exists a Z -cyclic patterned-starter $\text{Wh}(q^2)$ if and only if there exist $\frac{q+1}{2}$ pairs $(A_1, B_1), \dots, (A_{(q+1)/2}, B_{(q+1)/2})$ of elements of $\mathbb{Z}/q^2\mathbb{Z} - \{0\}$ whose configurations $[a_i, b_i; u_i, v_i]$, defined by (3), satisfy*

(P1) the sets $\{a_i, b_i\}, 1 \leq i \leq \frac{q+1}{2}$, partition $\mathcal{I}_q$

(P2) the sets $\{u_i, v_i\}, 1 \leq i \leq \frac{q+1}{2}$, partition $\mathcal{I}_q$

Proof. By [1], $\text{Wh}(q^2)$ exists if and only if both conditions (ϵ) and (ζ) hold. Therefore, to show that (P1) and (P2) are equivalent to the existence of $\text{Wh}(q^2)$, we must show that conditions (P1) and (P2) hold if and only if conditions (ϵ) and (ζ) hold.

In $\text{Wh}(q^2)$, each of the tables in the initial round are of the form $(rA_i, rB_i, -rA_i, -rB_i)$ where $T_i = (A_i, B_i, -A_i, -B_i)$ (i ranging from 1 to $\frac{q+1}{2}$) and each table T_i is turned into a set of $\frac{q-1}{2}$ tables through multiplying each of its entries by an element of R : $\{rT_i \mid r \in R\}$. In total, there are $\frac{q+1}{2} \cdot \frac{q-1}{2} = \frac{q^2-1}{4}$ tables, as desired.

(ϵ) is a condition about starters, which has 2 sub-conditions (α) and (β) that must hold. The first part of condition (ϵ) is (α) , which requires

$$\bigcup_{1 \leq i \leq \frac{q+1}{2}} \left(\bigcup_{r \in R} \{rA_i, -rA_i\} \cup \{rB_i, -rB_i\} \right) = \mathbb{Z}/q^2\mathbb{Z} - \{0\}$$

By Lemma 3.2, $\bigcup_{r \in R} \{rA_i, -rA_i\} = C(a_i)$ where $A_i \in C(a_i)$, and similarly for $\{rB_i, -rB_i\}$. Therefore, the expression simplifies to

$$\bigcup_{1 \leq i \leq \frac{q+1}{2}} C(a_i) \cup C(b_i)$$

By Lemma 3.3, the above union is equal to $\mathbb{Z}/q^2\mathbb{Z} - \{0\}$ if and only if $\{a_i, b_i\}$ partitions $\mathcal{I}_q$. Thus, $(\alpha) \iff (P1)$.

The second part of (ϵ) , which is (β) , requires the following:

$$\bigcup_{1 \leq i \leq \frac{q+1}{2}} \left(\bigcup_{r \in R} \{\pm 2rA_i\} \cup \{\pm 2rB_i\} \right) = \mathbb{Z}/q^2\mathbb{Z} - \{0\}$$

Elements $2rA_i, 2rB_i$ are in classes $C(a_i + \ell), C(b_i + \ell)$ respectively, with the infinity class getting mapped to itself: $C(\infty + \ell) = C(\infty)$. Since $j \mapsto j + \ell$ permutes $\mathcal{I}_q$, condition (β) follows from Lemma 3.3. Therefore, $(\beta) \iff (P1)$ and $(\epsilon) \iff (P1)$.

Consider condition (ζ) about 2-fold starters; like (ϵ) , it also contains 2 sub-conditions: (γ) and (δ) . Condition (γ) requires that the elements in the collection of pairs

$$\{rA_i, rB_i\}, \{rB_i, -rA_i\}, \{-rA_i, -rB_i\}, \{-rB_i, rA_i\}$$

over $1 \leq i \leq \frac{q+1}{2}$ and $r \in R$ are the elements of $\mathbb{Z}/q^2\mathbb{Z} - \{0\}$, each occurring twice. Due to the repetition within the pairs, this is equivalent to showing that the elements within $\{rA_i, -rA_i\} \cup \{rB_i, -rB_i\}$ over all $1 \leq i \leq \frac{q+1}{2}$ and $r \in R$ is equivalent to the elements of $\mathbb{Z}/q^2\mathbb{Z} - \{0\}$. This has been proven above; thus, $(\gamma) \iff (P1)$.

The second requirement of the 2-fold starter condition is about the differences between elements. For this collection of tables, the differences $\pm r(A_i - B_i)$ and $\pm r(A_i + B_i)$ each occur twice. Condition (δ) is thus equivalent to

$$\bigcup_{1 \leq i \leq \frac{q+1}{2}} \left(\bigcup_{r \in R} \{\pm r(A_i - B_i)\} \cup \{\pm r(A_i + B_i)\} \right) = \mathbb{Z}/q^2\mathbb{Z} - \{0\}$$

Let $A_i + B_i \in C(u_i)$ and $A_i - B_i \in C(v_i)$. By Lemma 3.2 the expression is equivalent to

$$\bigcup_{1 \leq i \leq \frac{q+1}{2}} C(u_i) \cup C(v_i)$$

By Lemma 3.3, the above union is equal to $\mathbb{Z}/q^2\mathbb{Z} - \{0\}$ if and only if $\{u_i, v_i\}$ partitions $\mathcal{I}_q$. Thus, $(\delta) \iff (P2)$ and ultimately (ζ) holds if and only if both (P1) and (P2) hold.

Since (ϵ) holds if and only if (P1) holds and (ζ) holds if and only if both (P1) and (P2) hold, conditions (P1) and (P2) are both sufficient and necessary for the existence of a Z-cyclic patterned starter $\text{Wh}(q^2)$. $\square$

Remark 3.5. Conditions (P1) and (P2) force $a_i \neq b_i$ and $u_i \neq v_i$ for every i . Configurations violating either are thus irrelevant to the problem and may be discarded. These cases are discarded when dealing with the hypergraph in Section 3.3.

3.2 The number of valid configurations

Proposition 3.4 focuses on the set of configurations that are *valid*; a configuration $[a, b; u, v]$ is valid if there exists a pair $(A, B) \in (\mathbb{Z}/q^2\mathbb{Z})^2$ that maps to $[a, b; u, v]$ via (3). Let

$$N(q) = \#\left\{ [a, b; u, v] : [a, b; u, v] \text{ is valid} \right\}.$$

This subsection focuses on evaluating the exact value of $N(q)$. Two lemmas are needed before calculating this value.

Lemma 3.6. *For any $0 \leq m \leq q - 1$, there exists a constant $C \not\equiv 0 \pmod{q}$ such that*

$$\log_w(1 + mq) \equiv Cm \pmod{q} \quad (5)$$

Proof. The case of $m = 0$ trivially holds. Let $U = \{1 + mq \pmod{q^2} : 0 \leq m \leq q - 1\}$. By Lemma 3.1 w is a primitive root modulo q so $w^{q-1} \equiv 1 \pmod{q}$. Therefore, let $w^{q-1} = cq + 1$ for some constant $c \neq 0$; $c \neq 0$ since w is a primitive root q^2 so $w^{q-1} \not\equiv 1 \pmod{q^2}$. By the binomial theorem, we have that

$$(w^{q-1})^k \equiv (cq + 1)^k \equiv kcq + 1 \pmod{q^2}$$

as all other terms are divisible by q^2 . When looking modulo q , the set $\{c, 2c, \dots, (q-1)c\}$ is equivalent to the set $\{1, 2, \dots, q-1\}$ as $\gcd(c, q) = 1$, so multiplying all elements of $\{1, 2, \dots, q-1\}$

by c is equivalent to permuting the elements of the set. Multiplying both sets by q show that the sets $\{q, 2q, \dots, (q-1)q\}$ and $\{cq, 2cq, \dots, (q-1)cq\}$ must be equivalent modulo q^2 . Therefore,

$$\begin{aligned} \{(w^{q-1})^k \pmod{q^2} : 0 \leq k \leq q-1\} &= \{(1+cq)^k \pmod{q^2} : 0 \leq k \leq q-1\} \\ &= \{1+cqk \pmod{q^2} : 0 \leq k \leq q-1\} \\ &= \{1+qk \pmod{q^2} : 0 \leq k \leq q-1\} \\ &= U \end{aligned}$$

Since the $m = 0$ case has been covered, the above reasoning shows that every element $1+mq \in U$, with $m \not\equiv 0 \pmod{q}$, can be represented as $(w^{q-1})^k \pmod{q^2}$ for some integer $k \not\equiv 0 \pmod{q}$. Finally,

$$\log_w(1+mq) = \log_w((w^{q-1})^k) = (q-1)k \equiv -k \pmod{q} \equiv Cm \pmod{q}$$

where $C = -km^{-1}$. □

Lemma 3.7 (independence of λ). *Fix $a, b \in \{0, \dots, q-1\}$. As A ranges over $C(a)$ and B over $C(b)$, the quantity $\lambda = BA^{-1} \pmod{q}$ takes every value in $\{1, 2, \dots, q-1\}$ an equal number of times.*

Proof. Let $A = w^{qk_1+a}$ and $B = w^{qk_2+b}$ with $0 \leq k_1, k_2 \leq q-2$. Then

$$\lambda = \frac{B}{A} = w^{q(k_2-k_1)+b-a} = (w^q)^{k_2-k_1} w^{b-a}.$$

By Lemma 3.1 w is a primitive root modulo q so $w^{q-1} \equiv 1$ and $w^q \equiv w \pmod{q}$. Therefore,

$$\lambda \equiv w^{k_2-k_1+b-a} \pmod{q}$$

Holding a, b, k_1 fixed and letting k_2 run over $\{0, 1, \dots, q-2\}$, the exponent runs over a complete residue system modulo $q-1$, so λ runs over $\{w^0, w^1, \dots, w^{q-2}\} \equiv \{1, 2, \dots, q-1\} \pmod{q}$. Since the exponent runs over a complete residue system for any k_1 , λ takes every value in $\{1, 2, \dots, q-1\}$ an equal number of times. □

Theorem 3.8. *For every prime $q \equiv 3 \pmod{4}$,*

$$N(q) = q^3 + q^2 + 1.$$

Every valid configuration is of one of the following six types, with the indicated multiplicities for each type:

	configuration	number
(T1)	$[\infty, \infty; \infty, \infty]$	1
(T2)	$[a, \infty; u, 2a - u \bmod q]$	q^2
(T3)	$[\infty, b; u, 2b - u \bmod q]$	q^2
(T4)	$[a, b; \infty, a + \ell \bmod q]$	q^2
(T5)	$[a, b; a + \ell \bmod q, \infty]$	q^2
(T6)	$[a, b; a + \log_w(1 + \lambda), a + \log_w(1 - \lambda)], \lambda \not\equiv 0, \pm 1$	$q^3 - 3q^2$

Proof. There are q^4 pairs $(A, B) \in (\mathbb{Z}/q^2\mathbb{Z})^2$, and each maps to a configuration by (3). We count the number of unique configurations $[a, b; u, v]$ that are generated over all q^4 pairs with casework. This map is not injective as multiple pairs can map to the same configuration.

Case 1: $q \mid A$ and $q \mid B$.

$a = b = \infty$, and since $A + B$ and $A - B$ are also divisible by q , $u = v = \infty$. All q^2 pairs map to the configuration $[\infty, \infty; \infty, \infty]$. This case contributes 1 configuration of type (T1).

Case 2: exactly one of A, B divisible by q .

By symmetry assume $q \nmid A$ and $q \mid B$. Then $b = \infty$ and $\log_w(A) \equiv a \pmod{q}$ with $a \in \{0, \dots, q-1\}$ by (4). Let $B = yq$ with $0 \leq y \leq q-1$. Since A is a unit in $\mathbb{Z}/q^2\mathbb{Z}$,

$$A + yq = A(1 + (yA^{-1})q),$$

where A^{-1} can be read modulo q since $(yA^{-1})q$ depends only on $yA^{-1} \pmod{q}$ when looking modulo q^2 . Lemma 3.6 therefore gives

$$u = \log_w(A + yq) \equiv \log_w(A) + CyA^{-1} \equiv a + CyA^{-1} \pmod{q}$$

for some constant C , and with the same constant,

$$v = \log_w(A - yq) \equiv a - CyA^{-1} \pmod{q}.$$

Combining the 2 gives $u + v \equiv 2a \pmod{q}$, and every configuration in this case has the form $[a, \infty; u, 2a - u]$. Conversely, given any a and any u , choose $A \in C(a)$ and let $y \equiv C^{-1}(u - a)A \pmod{q}$; this creates the configuration $[a, \infty; u, 2a - u]$. a and q can be independently chosen from q options, thus giving q^2 configurations of type (T2). Letting $q \mid A$ and $q \nmid B$ gives q^2 configurations of type (T3) by symmetry.

Case 3: $q \nmid A$ and $q \nmid B$.

Both A and B are units. Let $\lambda \equiv BA^{-1} \pmod{q}$, so $\lambda \in \{1, 2, \dots, q-1\}$. Then

$$u = \log_w(A + B) \equiv \log_w(A) + \log_w\left(1 + \frac{B}{A}\right) \equiv a + \log_w(1 + \lambda) \pmod{q},$$

$$v = \log_w(A - B) \equiv \log_w(A) + \log_w\left(1 - \frac{B}{A}\right) \equiv a + \log_w(1 - \lambda) \pmod{q},$$

If $\lambda \equiv -1 \pmod{q}$ then $q \mid A + B$ so $u = \infty$, while $v \equiv a + \log_w(2) = a + \ell$. By Lemma 3.7 the value $\lambda \equiv -1$ is attainable for every choice of a and b , so this subcase yields q^2 configurations $[a, b; \infty, a + \ell]$ of type (T4). Symmetrically, $\lambda \equiv 1$ gives $v = \infty$, $u \equiv a + \ell$, and q^2 configurations of type (T5).

If $\lambda \not\equiv \pm 1$ there are $q-3$ possible values of λ : $\lambda \in \{2, 3, \dots, q-2\}$. By Lemma 3.7, λ may be chosen independently with fixed values for a and b , so a , b and λ range independently over q , q and $q-3$ values respectively, giving $q^2(q-3) = q^3 - 3q^2$ configurations of type (T6).

Adding the number of configurations for all 6 types results in

$$N(q) = 1 + 2q^2 + 2q^2 + q^3 - 3q^2 = q^3 + q^2 + 1$$

□

3.3 The hypergraph $H(q)$

This section will focus on constructing the hypergraph that will be used for the matching problem discussed in Proposition 3.4.

Definition 3.9. Let V_{AB} and V_{UV} be two disjoint copies of $\mathcal{I}_q$, where we refer to a vertex $v \in V_{AB}$ as v_{AB} (likewise for V_{UV}). Let $V = V_{AB} \cup V_{UV}$ so that $n = |V| = 2(q+1)$. Let $H(q) = (V, E)$ be the hypergraph whose edges are

$$e = \{a_{AB}, b_{AB}, u_{UV}, v_{UV}\}$$

for each valid configuration $[a, b; u, v]$ with $a \neq b$ and $u \neq v$.

By Remark 3.5, we discard the configurations with $a = b$ or $u = v$, as they can never appear in a collection of quadruples satisfying (P1) and (P2): type (T1) has $a = b$; types (T2) and (T3) have $u = v$ when $u \equiv a$ and $u \equiv b$ respectively, which is q cases each; types (T4) and (T5) have $u \neq v$ and $a = b$ in q cases each; and type (T6) has $a = b$ in $q(q-3)$ cases and never has $u = v$. Therefore,

$$|E| = N(q) - (q(q-3) + 4q + 1) = q^3 - q \tag{6}$$

$H(q)$ is 4-uniform since every hyperedge has 4 vertices; furthermore, every edge has exactly two vertices in each of V_{AB} and V_{UV} . As a result, any matching in $H(q)$ covers equally many vertices on each part of the hypergraph, and Proposition 3.4 can be reframed as the following.

Corollary 3.10. *A Z -cyclic patterned-starter $\text{Wh}(q^2)$ exists if and only if $H(q)$ has a perfect matching, i.e. a set of $\frac{q+1}{2}$ pairwise disjoint edges.*

The next lemma proves that the condition required to use the hypergraph matching theorem used in Section 3.4 holds for H .

Lemma 3.11. *Let $D = D(q) = 2q^2$, the degree $d(x)$ be the number of hyperedges that contain $x \in V$, and the codegree $d(x, y)$ be the number of hyperedges that contain both $x, y \in V$. Then,*

$$d(x) = D + O(q) \quad \forall x \in V, \quad d(x, y) \leq 4q + O(1) \quad \forall x, y \in V, x \neq y.$$

In particular $d(x) = (1 + O(q^{-1}))D$ and $d(x, y) = O(D/q) = o(D)$.

Proof. We count using the type classification in Theorem 3.8. Throughout the proof, j denotes a finite index, j_{AB} denotes a vertex in V_{AB} , and j_{UV} denotes a vertex in V_{UV} . The first part of the proof counts the degrees of vertices in V_{AB} , the second part counts the degrees of vertices in V_{UV} and the third part counts the codegrees of vertices in the hypergraph.

Degrees in V_{AB}

For $x = j_{AB}$ with j finite (not equal to ∞), the edges containing x come from type (T6) configurations with $a = j$ or $b = j$; there are $(q-1)(q-3)$ of each, since the other player index can be any of the other $j-1$ finite indices and λ has $q-3$ values. Types (T2) through (T5) contribute $O(q)$ edges in total, since for each type there is only one degree of freedom for a configuration $[a, b; u, v]$. Therefore, $d(j_{AB}) = 2(q-1)(q-3) + O(q) = 2q^2 + O(q)$. For $x = \infty_{AB}$ (vertex that corresponds to the ∞ set), only types (T2) and (T3) qualify, giving $q(q-1)$ edges each, so $d(\infty_{AB}) = 2q^2 - 2q = 2q^2 + O(q)$.

Degrees in V_{UV}

For $x = j_{UV}$ with j finite, type (T6) contributes the edges with $u = j$; for each of the $q-3$ values of λ the index a is determined by $a \equiv j - \log_w(1 + \lambda)$ and b can be chosen freely, giving $(q-3)(q-1)$ edges and symmetrically $(q-3)(q-1)$ edges with $v = j$. Types (T2) through (T5) contribute $O(q)$ edges. Therefore, $d(j_{UV}) = 2q^2 + O(q)$. For $x = \infty_{UV}$, only types (T4) and (T5) contribute edges, each with a, b free and distinct, so $d(\infty_{UV}) = 2q(q-1) = 2q^2 - 2q = 2q^2 + O(q)$. Thus, the degree of any vertex in the graph is $D + O(q)$.

Codegrees

Let x, y be vertices with $x \neq y$. There are 5 possible cases:

1. Both indices are finite and from V_{AB}
2. Both indices are finite and from V_{UV}
3. Both indices are finite; one is from V_{AB} and the other is from V_{UV}
4. One of the indices is ∞_{AB} and the other is finite
5. One of the indices is ∞_{UV} and the other is finite

For case 1, if $x = j_{AB}$ and $y = j'_{AB}$ with j, j' finite, an edge containing both has $\{a, b\} = \{j, j'\}$. (T4) and (T5) each contribute 2 configurations, and (T6) contributes $2(q-3)$ since λ can be freely chosen from $\lambda-3$ values. Therefore, the codegree is at most $2(q-3) + 4 = 2(q-1)$.

For case 2, if $x = j_{UV}$ and $y = j'_{UV}$ with j, j' finite then an edge containing both must have $\{u, v\} = \{j, j'\}$. Types (T2) and (T3) each contribute $O(1)$ to the codegree as fixing $\{j, j'\} = \{u, 2a-u\}$ or $\{j, j'\} = \{u, 2b-u\}$ leaves no degrees of freedom. If the edge containing x, y was type (T6), then $j - j' = \log_w((1 + \lambda)(1 - \lambda)^{-1})$, so the difference $j - j'$ would determine $(1 + \lambda)(1 - \lambda)^{-1} \pmod{q}$ and hence λ . This in turn would determine the value of a , letting b be freely chosen for a codegree of $2(q-1)$.

For case 3, if $x = j_{AB}$ and $y = j'_{UV}$, then x, y can appear as vertices in any of the configuration types from (T2) through (T6). (T2) and (T3) each contribute a codegree of $O(1)$, as fixing a and u or a and $2a-u$ determines the configuration (likewise for b and u or b and $2b-u$). (T4) and (T5) contribute $O(1)$ to the codegree as well for the same reason. For (T6), there are 4 ways of assigning x to a or b and y to u or v , each of which contributes $q-1$ to the codegree. Therefore, pairs of vertices in case 3 have a codegree of $4q + O(1)$.

For case 4, if one of the vertices is ∞_{AB} and the finite index is fixed then the edge is of type (T2) or (T3) and the codegree is $2(q-1)$. For case 5, if one of the vertices is ∞_{UV} and the finite index is fixed then the edge containing both vertices is of type (T4) or (T5); the codegree is $2(q-1)$. As desired, the codegree of any 2 distinct vertices $d(x, y) \leq 4q + O(1)$ $\square$

3.4 Applying Pippenger's theorem

Theorem 3.12 (Frankl–Rödl–Pippenger [13, 14, 12]). *For every integer $k \geq 2$ and every $\varepsilon > 0$ there exists $\delta > 0$ such that the following holds. Let H be a k -uniform hypergraph on n vertices, and suppose there is a number D with*

1. $(1 - \delta)D \leq d(x) \leq (1 + \delta)D$ for all $x \in V(H)$, and
2. $d(x, y) \leq \delta D$ for all distinct $x, y \in V(H)$.

Then H contains a matching (a collection of pairwise disjoint edges) covering all but at most εn of the vertices.

Since we are working asymptotically with degrees and codegrees, hypotheses (1) and (2) are equivalent to the condition $d(x, y) = o(d(x))$.

Theorem 3.13. $\forall \varepsilon > 0$ there is a $q_0(\varepsilon)$ such that for every prime $q \equiv 3 \pmod{4}$ with $q \geq q_0(\varepsilon)$ there exist

$$m \geq \frac{q+1}{2} - o(q)$$

valid configurations $[a_1, b_1; u_1, v_1], \dots, [a_m, b_m; u_m, v_m]$ such that the sets $\{a_i, b_i\}$ are pairwise disjoint and the sets $\{u_i, v_i\}$ are pairwise disjoint. Equivalently, all but at most $2\varepsilon(q+1)$ of the coset indices in the two copies of $\mathcal{I}_q$ are covered.

Proof. Apply Theorem 3.12 to $H = H(q)$ with $k = 4$, $n = 2(q+1)$ and $D = 2q^2$. Given $\varepsilon > 0$, let $\delta > 0$ be as stated by the theorem. By Lemma 3.11 there exists a q_0 such that for $q \geq q_0$ we have $(1 - \delta)D \leq d(x) \leq (1 + \delta)D$ for all x and $d(x, y) \leq 4q + O(1) \leq \delta D$ for all distinct x, y . The former inequality holds since $d(x) = D + o(D)$ and the latter holds because $\lim_{q \rightarrow \infty} q/D = 1/2q = 0$.

Theorem 3.12 yields a matching M covering at least $(1 - \varepsilon)n = (1 - \varepsilon)(2q + 2)$ vertices. Since $H(q)$ is 4-uniform and M consists of pairwise disjoint edges,

$$|M| \geq \frac{(1 - \varepsilon)(2q + 2)}{4} = \frac{q+1}{2}(1 - \varepsilon) = \frac{q+1}{2} - o(q).$$

Each edge of M is, by Definition 3.9, a valid configuration, and the edges are disjoint since the sets $\{a_i, b_i\}$ and $\{u_i, v_i\}$ are pairwise disjoint. $\square$

Theorem 3.13 says that as $q \rightarrow \infty$ one can always find a number a collection of quadruples arbitrarily close to the required $\frac{q+1}{2}$, whose player indices and difference indices each cover a proportion of $\mathcal{I}_q$ arbitrarily close to 1.

3.5 What remains

The gap between Theorem 3.13 and Conjecture 1.3 is the difference between a near-perfect and a perfect matching in $H(q)$. Conditions (P1) and (P2) are exact partition conditions that require all of the $2(q+1)$ vertices of $H(q)$ to be covered, whereas Pippenger's Theorem 3.12 leaves a set $L \subseteq V$ of size $|L| \leq \varepsilon n$ uncovered. The following is a problem that if solved, would bridge the work done in this paper to proving Conjecture 1.3.

Remaining problem. Show that for all sufficiently large primes $q \equiv 3 \pmod{4}$, the near-perfect matching created through Theorem 3.13 can be transformed into a perfect matching of $H(q)$.

Any completion algorithm which covers the residual $|L|$ vertices will have to adhere to strict algebraic constraints, as the set of edges is quite precise; the hypergraph $H(q)$ contains 6 explicit types of edges, and the behaviors of the differences $A \pm B$ place further arithmetic constraints. Any completion argument must work within these constraints to perfectly cover the hypergraph.

Finally, we note that even a full solution of the remaining problem would prove Conjecture 1.3 only for $q \geq q_0$, leaving finitely many primes to be checked between 5000 and the threshold q_0 . Obtaining an effective and small threshold q_0 is also nontrivial and could be the focus of future work.

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